

Unity Projection Mapping SDK - Quick Start Guide

Welcome to the Projection Mapping SDK for Unity! This package provides high-performance, multithreaded native C++ projection mapping capabilities directly inside Unity without locking the main thread.

Installation

1. Open the Unity Package Manager (Window > Package Manager).
2. Click the + button in the top left and select Add package from disk...
3. Navigate to this folder (com.viitorx.pmsdk) and select the package.json file.
4. Unity will automatically import the SDK and its native binaries.

Getting Started

1. The Interactive Setup Wizard

The easiest way to get started is to use the Guided Setup Wizard: 1. In the top menu bar, click Tools > Projection Mapping > Setup Wizard. 2. Follow the on-screen steps to configure your Projector and physical Camera. 3. The wizard will automatically generate the DeformationField, WarpNode, and Mesh for your scene.

2. Manual C# Scripting

If you want to manually interact with the SDK, you can use the C# wrappers:

```
```csharp using PMSDK;

public class ProjectionMapper : MonoBehaviour { void Start() { // 1. Create a Warp Node IntPtr warpNode = NativeBindings.CreateWarpNode("MyProjector");

// 2. Load a 3D Mesh
IntPtr mesh = NativeBindings.LoadMesh("path/to/mesh.obj");

// 3. Process the mapping (runs entirely in native C++)
NativeBindings.ProcessMesh(warpNode, mesh);
}
}
```
```

Advanced: GrayCode Calibration

This SDK supports automatic GrayCode generation and decoding to automatically triangulate your physical projection surface using a camera! 1. Open the Calibration tools in the Unity menu. 2. Select your Camera (the SDK connects directly to it via C++ cv::VideoCapture to ensure zero lag in Unity). 3. Click Start Calibration. The projector will flash GrayCode patterns, and the system will generate a pixel-perfect 3D mesh of your real-world surface!